AI-Assisted Creative Expression: a Case for Automatic Lineart Colorization

Yliess Hati

January 27, 2023

Contents

Abstract **1**

Aknowledgements **3**

I Context **5**

1 Introduction **7**
1.1 Motivations 8
1.2 Problem Statement 8
1.3 Contributions 9
1.4 Concerns 9
1.5 Outline 10

2 Background **11**
2.1 History of Artificial Intelligence 11
2.2 Neural Networks 11
2.3 Autoencoders 11
2.4 Variational Autoencoders 11
2.5 Generative Adversarial Networks 11
2.6 Denoising Diffusion Models 11

3 Methodology **13**
3.1 Implementation 13
3.2 Objective Evaluation 13
3.3 Subjective Evaluation 13
3.4 Reproducibility 13

II Core **15**

4 Contrib I (Find Catchy Explicit Name) **17**
4.1 State of the Art 17
4.2 Method 17
4.3 Setup 17
4.4 Results 17
4.5 Summary 17

5 Contrib II (Find Catchy Explicit Name) 19
5.1 State of the Art 19
5.2 Method 19
5.3 Setup 19
5.4 Results 19
5.5 Summary 19

6 Contrib III (Find Catchy Explicit Name) 21
6.1 State of the Art 21
6.2 Method 21
6.3 Setup 21
6.4 Results 21
6.5 Summary 21

7 Contrib IV (Find Catchy Explicit Name) 23
7.1 State of the Art 23
7.2 Method 23
7.3 Setup 23
7.4 Results 23
7.5 Summary 23

III Reflection 25

8 Ethical and Societal Impact 27

9 Conclusion 29

References 31

[29] Zhang, R., Zhu, J.-Y., Isola, P., Geng, X., Lin, A.S., Yu, T. and Efros, A.A. 2017. Real-time user-guided image colorization with learned deep priors. *ACM Trans. Graph.* 36, 4 (Jul. 2017). DOI:https://doi.org/10.1145/3072959.3073703[23].

[13] Kim, H., Jhoo, H.Y., Park, E. and Yoo, S. 2019. Tag2Pix: Line art colorization using text tag with SECat and changing loss[9]. *2019 IEEE/CVF international conference on computer vision (ICCV)* (2019), 9055–9064.

[14] Kingma, D.P. and Welling, M. 2013. Auto-encoding variational bayes. *arXiv preprint arXiv:1312.6114*. (2013).

[15] Le Cun, Y. 2019. *Quand la machine apprend: La révolution des neurones artificiels et de l'apprentissage profond*[10]. Odile Jacob.

[16] Liu, Y., Qin, Z., Wan, T. and Luo, Z. 2018. Auto-painter: Cartoon image generation from sketch by using conditional wasserstein generative adversarial networks. *Neurocomputing*. 311, (2018), 78–87. DOI:https://doi.org/10.1016/j.neucom.2018.05.045[11].

[17] Mumford, M., Medeiros, K. and Partlow, P. 2012. Creative thinking: Processes, strategies, and knowledge. *The Journal of Creative Behavior*. 46, (Mar. 2012). DOI:https://doi.org/10.1002/jocb.003[12].

[18] Newell, A., Shaw, J.C. and Simon, H.A. 1959. *The processes of creative thinking*[13]. RAND Corporation.

[19] Paintman: *http://www.retasstudio.net/products/paintman/*[14]. Accessed: 2023-01-26.

[20] Pelica paint: 2017. *https://petalica-paint.pixiv.dev/index_en.html*[15]. Accessed: 2023-01-26.

[21] Photoshop: *https://www.adobe.com/products/photoshop.html*[16]. Accessed: 2023-01-26.

[22] Rombach, R., Blattmann, A., Lorenz, D., Esser, P. and Ommer, B. 2021. High-resolution image synthesis with latent diffusion models[17].

[23] Rumelhart, D.E., Hinton, G.E. and Williams, R.J. 1986. Learning representations by back-propagating errors. *Nature*. 323, 6088 (Oct. 1986), 533–536. DOI:https://doi.org/10.1038/323533a0[18].

[24] Saito, M. and Matsui, Y. 2015. Illustration2Vec: A semantic vector representation of illustrations[19]. *SIGGRAPH asia 2015 technical briefs* (New York, NY, USA, 2015), 5:1–5:4.

[25] Sangkloy, P., Lu, J., Fang, C., Yu, F. and Hays, J. 2017. Scribbler: Controlling deep image synthesis with sketch and color[20]. *2017 IEEE conference on computer vision and pattern recognition, CVPR 2017, honolulu, HI, USA, july 21-26, 2017* (2017), 6836–6845.

[26] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł. and Polosukhin, I. 2017. Attention is all you need. *Advances in neural information processing systems*. 30, (2017).

[27] Wolf, M.J., Miller, K. and Grodzinsky, F.S. 2017. Why we should have seen that coming: Comments on microsoft's tay "experiment," and wider implications. *SIGCAS Comput. Soc.* 47, 3 (Sep. 2017), 54–64. DOI:https://doi.org/10.1145/3144592.3144598[21].

[28] Zhang, L., Ji, Y., Lin, X. and Liu, C. 2017. Style transfer for anime sketches with enhanced residual u-net and auxiliary classifier GAN[22]. *2017 4th IAPR asian conference on pattern recognition (ACPR)* (2017), 506–511.

Abstract

References

[1] Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J.D., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A., et al. 2020. Language models are few-shot learners. *Advances in neural information processing systems*. 33, (2020), 1877–1901.

[2] CHATGPT: Optimizing language models for dialogue: 2023. *https://openai.com/blog/chatgpt/*[1]. Accessed: 2023-01-26.

[3] Ci, Y., Ma, X., Wang, Z., Li, H. and Luo, Z. 2018. User-guided deep anime line art colorization with conditional adversarial networks[2]. *Proceedings of the 26th ACM international conference on multimedia* (New York, NY, USA, 2018), 1536–1544.

[4] Clip studio PAINT: *https://www.clipstudio.net/*[3]. Accessed: 2023-01-26.

[5] Frans, K. 2017. Outline colorization through tandem adversarial networks[4]. *CoRR*. abs/1704.08834, (2017).

[6] Furusawa, C., Hiroshiba, K., Ogaki, K. and Odagiri, Y. 2017. Comicolorization: Semi-automatic manga colorization[5]. *SIGGRAPH asia 2017 technical briefs* (New York, NY, USA, 2017).

[7] Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A. and Bengio, Y. 2014. Generative adversarial nets[6]. *Advances in neural information processing systems* (2014).

[8] Hati, Y., Jouet, G., Rousseaux, F. and Duhart, C. 2019. PaintsTorch: A user-guided anime line art colorization tool with double generator conditional adversarial network[7]. *European conference on visual media production* (New York, NY, USA, 2019).

[9] Hati, Y., Thevenin, V., Nolot, F., Rousseaux, F. and Duhart, C. 2022. StencilTorch: An iterative and user-guided framework for anime lineart colorization. (2022).

[10] Hensman, P. and Aizawa, K. 2017. cGAN-based manga colorization using a single training image[8]. *2017 14th IAPR international conference on document analysis and recognition (ICDAR)* (Los Alamitos, CA, USA, Nov. 2017), 72–77.

[11] Ho, J., Jain, A. and Abbeel, P. 2020. Denoising diffusion probabilistic models. *Advances in Neural Information Processing Systems*. 33, (2020), 6840–6851.

[12] Kandinsky, W. and Sadleir, M. 1977. *Concerning the spiritual in art*. Dover Publications.

Aknowledgements

Chapter 9

Conclusion

Part I

Context

Chapter 8

Ethical and Societal Impact

Chapter 1

Introduction

Humans possess the ability to perceive and understand the world allowing us to accomplish a wide range of complex tasks through the combination of visual recognition, scene understanding, and communication. The ability to quickly and accurately extract information from a single image is a testament to the complexity and sophistication of the human brain and is often taken for granted. One of the Artificial Intelligence (AI) field's ultimate goals is to empower computers with such human-like abilities, one of them being creativity, being able to produce something original and worthwhile [17].

Computational creativity is the field at the intersection of AI, cognitive psychology, philosophy, and art, which aims at understanding, simulating, replicating, or in some cases enhancing human creativity. One definition of computational creativity [18] is the ability to produce something that is novel and useful, demands that we reject common beliefs, results from intense motivation and persistence, or comes from clarifying a vague problem. Top-down approaches to this definition use a mix of explicit formulations of recipes and randomness such as procedural generation. On the opposite, bottom-up approaches use Artificial Neural Networks (ANNs) to learn patterns and heuristics from large datasets to enable non-linear generation.

We, as a species, are currently witnessing the beginning of a new era where the gap between machines and humans is starting to blur. Current breakthroughs in the field of AI, more specifically in Deep Learning (DL), are giving computers the ability to perceive and understand our world, but also to interact with our environment using natural interactions such as speech and natural language. ANNs, once mocked by the AI community [15], are now trainable using Gradient Descent (GD) [23] thanks to the massive availability of data and the processing power of modern hardware accelerators such as Graphical Processing Units (GPUs), Tensor Processing Units (TPUs), and Neural Processing Units (NPUs).

Neural Networks (NNs), those trainable general function approximators, gave rise to the field of generative NN. Specialized DL architectures such as Variational Autoencoders (VAEs) [14], Generative Adversarial Networks (GANs) [7], Denoising Diffusion Models (DDMs) [11], and Large Language Models (LLMs) [1, 26] are used to generate artifacts such as text, audio, images, and videos of

unprecedented quality and complexity.

This dissertation aims at exploring how one could train and use generative NN to create AI-powered tools capable of enhancing human creative expression. The task of automatic lineart colorization act as the example case used to illustrate this process throughout the entire thesis.

1.1 Motivations

Lineart colorization is an essential aspect of the work of artists, illustrators, and animators. The task of manually coloring lineart can be time-consuming, repetitive, and exhausting, particularly in the animation industry, where every frame of an animated product must be colored and shaded. This process is typically done using image editing software such as Photoshop [21], Clip Studio PAINT [4], and PaintMan [19]. Automating the colorization process can greatly improve the workflow of these creative professionals and has the potential to lower the barrier for newcomers and amateurs. Such a system was integrated into Clip Studio PAINT [4], demonstrating the growing significance of automatic colorization in the field.

The most common digital illustration process can be broken down into four distinct stages: sketching, inking, coloring, and post-processing. As demonstrated by the work of Kandinsky [12], the colorization process can greatly impact the overall meaning of a piece of art through the introduction of various color schemes, shading, and textures. These elements of the coloring process present significant challenges for the Computer Vision (CV) task of automatic lineart colorization, particularly in comparison to its grayscale counterpart [6, 10, 29]. Without the added semantic information provided by textures and shadows, inferring materials and 3D shapes from black and white linearts is difficult. They can only be deduced from silhouettes.

1.2 Problem Statement

One major challenge of automatic lineart colorization is the availability of qualitative public datasets. Illustrations do not always come with their corresponding lineart. The few dataset available for the task are lacking consistency in the quality of the illustrations, gathering images from different types, mediums and styles. For those reasons, online scrapping and synthetic lineart extraction is the method of choice for many of the contributions in the field [3, 29].

Previous works in automatic lineart colorization are based on the GAN [7] architecture. They can generate unperfect but high-quality illustrations in a quasi realtime setting. They achieve user control and guidance via different means, color hints [3, 5, 16, 20, 25], style transfer [28], tagging [13], and more recently natural language [11]. One common pattern in these methods is the use of a feature extractor such as Illustration2Vec [24] allowing to compensate for the lack of semantic descriptors by injecting its feature vector into the models.

Part III

Reflection

1.3 Contributions

This work focuses on the use of color hints in the form of user strokes as it fits the natural digital artist workflow and does not involve learning and mastering a new skill. While previous works offers improving quality compared to classical CV techniques, they are still subject to noisy training data, artifacts, a lack of variety, and a lack of fidelity in the user intent. In this dissertation we explore the importance of a clean, qualitative and consistent dataset. We investigate how to better capture the user intent via natural artistic controls and how to reflect them into the generated model artifact while preserving or improving its quality. We also look at how the creative process can be transformed into a dynamic iterative workflow where the user collaborates with the machine to refine and carry out variations of his artwork.

Here is a brief enumeration of this thesis's contributions:

- We present a recipe for curating datasets for the task of automatic lineart colorization [8, 9]
- We introduce three generative models:
  - PaintsTorch [8], a double GAN generator that improved generation quality compared to previous work while allowing realtime interaction with the user.
  - StencilTorch [9], an upgrade upon PaintsTorch, shifting the colorization problem to in-painting allowing for human collaboration to emerge as a natural workflow where the input of a first pass becomes the potential input for a second.
  - StablePaint, an exploration of DDM for bringing more variety into the generated outputs allowing for variation exploration and conserving the iterative workflow introduced by StencilTorch for the cost of inference speed.
- We offer an advised reflection on current generative AI ethical and societal impact on our society.

1.4 Concerns

Recent advances in generative AI for text, image, audio, and video synthesis are raising important ethical and societal concerns for our society, especially because of its availability and ease of use. Models such as Stable Diffusion [22] and more recently Chat-GPT [2] are disturbing our common beliefs and relation with copyright, creativity, the distribution of fake information and so on.

One of the main issues with generative AI is the potential for model fabulation. Generative models can create entirely new, synthetic data that is indistinguishable from real data. This can lead to the dissemination of false information and the manipulation of public opinion. Additionally, there are ambiguities surrounding the ownership and copyright of the generated content, as it is unclear who holds the rights to the generated images and videos. Training data is often obtained via online scrapping and thus copyright ownership is not propagated. This is especially true for commercial applications.

Another important concern is the potential for biases and discrimination. These

models are trained on large amounts of data, and if the data is not diverse or representative enough, the model may perpetuate or even amplify existing biases. The Microsoft Tay Twitter bot [27]] scandal is an outcome of such a phenomenon. This initially innocent chatbot has been easily turned into a racist bot perpetuating hate speech. The task was made easier because of the inherently biased dataset it was trained on.

In this work, we are committed to addressing and raising awareness for these concerns. The illustrations used for training our models and for our experiments are only used for educational and research purposes. We only provide recipes for reproducibility and do not distribute the dataset nor the weights resulting from model training, only the code. We hope this will not ensure that our work is used ethically and responsibly but limit its potential misuse.

1.5 Outline

The first part of this thesis (chapters 1-3) provides context to the recent advances in generative AI and introduces the CV task of user-guided automatic linear colorization, its challenges, and our contributions to the field. It then provides additional background, from DL first principles to current architectures used in modern generative NN, and introduces the methodology used throughout the entire document. This part should be accessible to the majority, experts and non-experts, and serve as an introduction to the field.

The second part (chapters 4-7) presents our contributions, some of which have previously been presented in [8, 9]. It introduces into detail our recipe for sourcing and curation of consistent and qualitative datasets for automatic lineart colorization, PaintsTorch [8] our first double generator GAN conditioned on user strokes, StencilTorch [9] our in-painting reformulation introducing the use of masks to allow the emergence of iterative workflow and collaboration with the machine, and finally StablePaint, an exploration of the use of DDM models for variations qualitative exploration.

The third and final part (chapters 7-8) offers a detailed reflection on this thesis's contributions and more generally about the field of generative AI ethical and societal impact, identifies the remaining challenges and discusses future work.

The code base for the experiments and contributions are publicly available on GitHub at https://github.com/yliess86

Chapter 7

Contrib IV (Find Catchy Explicit Name)

7.1 State of the Art

7.2 Method

7.3 Setup

7.4 Results

7.5 Summary

Chapter 2

Background

2.1 History of Artificial Intelligence

2.2 Neural Networks

2.3 Autoencoders

2.4 Variational Autoencoders

2.5 Generative Adversarial Networks

2.6 Denoising Diffusion Models

Chapter 6

Contrib III (Find Catchy Explicit Name)

6.1 State of the Art

6.2 Method

6.3 Setup

6.4 Results

6.5 Summary

Chapter 3

Methodology

3.1 Implementation

3.2 Objective Evaluation

3.3 Subjective Evaluation

3.4 Reproducibility

Chapter 5

Contrib II (Find Catchy Explicit Name)

5.1 State of the Art

5.2 Method

5.3 Setup

5.4 Results

5.5 Summary

Part II

Core

Chapter 4

Contrib I (Find Catchy Explicit Name)

4.1 State of the Art

4.2 Method

4.3 Setup

4.4 Results

4.5 Summary